

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 12: Reproduction in Plants

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	12 — Reproduction in Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors weaponise the deepest confusions: pollen-transfer vs gamete-fusion, ovule->seed vs ovary->fruit vs zygote->embryo, clone-uniformity vs seed-variation, spore (no fusion) vs seed (post-fusion), agent-matched adaptations, and the precise order of life-cycle events under experimental manipulation. Do not write on this paper — mark answers on your answer sheet.

1. A horticulturist propagates a seedless grape vine entirely from cuttings for forty years. A visiting scientist claims this means the variety can never produce offspring with any new trait through this method. The scientist's claim is best judged as:

- (A) unsound, because repeated cuttings gradually accumulate new traits over generations
- (B) unsound, because each cutting secretly involves pollen from a second parent
- (C) sound, but only because grapes happen to be seedless
- (D) sound, because cuttings copy a single parent's genetic material exactly, so no reshuffling of traits occurs

2. A field grown from cuttings of one elite parent and a neighbouring field grown from that parent's seeds are both exposed to a brand-new pest. The cutting field is wiped out almost uniformly; the seed field loses only part of its plants. The deepest reason the seed field fared better is that:

- (A) the seed plants were younger and pests prefer older plants
- (B) the seed plants carry varied combinations of traits, so some happened to resist the pest while clones shared one weakness

- (C) the cutting plants had used up their stored food and starved
- (D) seeds release a chemical that repels all pests automatically

3. A student stores two potato pieces: one with three eyes and one from which all eyes were carefully gouged out, then plants both in identical moist soil. After three weeks only one piece has sprouted shoots. The piece that failed lacked:

- (A) the stored food, which the gouging removed entirely
- (B) the buds, from which the new shoots would have grown
- (C) the roots, which only sprout from eyeless surfaces
- (D) the ovules, which the eyes normally contain

4. A teacher lays a single Bryophyllum leaf on damp soil and a single fern frond bearing brown patches on adjacent damp soil. Weeks later both have produced young plants, yet a biologist insists these two events are fundamentally unlike. The sharpest statement of the difference is:

- (A) the leaf used spores while the frond used buds on its margin
- (B) both used seeds, but the leaf's seeds were hidden inside it
- (C) the leaf's plantlets grow from buds (vegetative propagation), while the frond's young plants arise from spores released from spore cases
- (D) the leaf reproduced sexually and the frond reproduced by cuttings

5. Under a microscope, a yeast cell shows a small swelling that narrows at its base and finally pinches free, while in a separate dish a Spirogyra filament has snapped into several pieces that each keep growing. A student labels both 'fragmentation'. The correct correction is that:

- (A) the yeast event is budding (one outgrowth separates), and only the Spirogyra event is fragmentation (the body breaks into pieces)
- (B) both are correctly fragmentation, since something separates in each case
- (C) the yeast is fragmentation and the Spirogyra is budding
- (D) both are spore formation because a parent cell is involved

6. A mushroom, a moss, and a fern are grouped in a lesson, and beside them sits a potato. A student is asked which one does NOT share the grouping feature. The correct answer and reason is:

- (A) the moss, because mosses alone make seeds, unlike the spore-forming others
- (B) the fern, because ferns alone reproduce sexually by flowers
- (C) the potato, because the other three reproduce by spores while the potato sprouts vegetatively from buds in its tuber
- (D) the mushroom, because fungi cannot reproduce at all without a host

7. Spores survive a long dry spell and germinate only after the first rains, whereas a delicate water alga must keep dividing to persist. The chief survival value the spore offers that the alga's method does not is that the spore:

- (A) combines with a second spore so the pair survives drought together
- (B) can wait, protected in its case, through harsh conditions and grow only when moisture returns
- (C) is heavier, so it sinks and stays moist underground
- (D) germinates the instant it forms, before any dry spell can harm it

8. A gardener wants a winter supply of new money plants and a separate batch of varied tomato plants for breeding trials. The methods best matched to each goal are:

- (A) sowing seeds for both, to keep each batch uniform
- (B) stem cuttings for both, to maximise variation in each
- (C) stem cuttings for the money plants and sowing seeds for the tomatoes
- (D) scattering spores for the money plants and cuttings for the tomatoes

9. In grafting, a shoot of a prized apple variety is joined to the rooted stem of a hardy rootstock. A nursery owner argues this lets him sell apples that are a brand-new blend of the two plants' fruit traits. His reasoning is:

- (A) right, because grafting fuses the gametes of both plants
- (B) right, because the rootstock's pollen reaches the shoot's flowers
- (C) wrong, because the grafted shoot keeps the prized variety's own traits rather than blending with the rootstock
- (D) wrong, because grafting actually produces spores instead of fruit

10. A child reasons: 'Onion grows from a bulb, ginger from an underground stem, and a fern from spores — so none of these can be called asexual reproduction, since no buds are involved.' The flaw in this reasoning is that:

- (A) all three are asexual; bulbs and underground stems are vegetative routes and spores are asexual units, none involving fusion of gametes
- (B) only the fern is asexual; bulbs and stems are forms of sexual reproduction
- (C) asexual reproduction always requires buds, so the child is correct
- (D) onion and ginger are sexual because they grow underground in the dark

11. Two cuttings are taken from the same parent rose. Cutting A is planted in soil, cutting B is left in a glass of water; both grow into rooted plants. A student concludes that the two new roses must differ genetically because they grew in different media. This conclusion is:

- (A) right, because the water cutting absorbed traits from the water
- (B) right, because soil and water trigger different parent genes permanently
- (C) wrong, because the soil cutting is actually grown from a seed
- (D) wrong, because both are clones of the same single parent regardless of the medium they rooted in

12. A diagram of one flower labels four parts: P produces pollen, Q is a slender stalk holding P aloft, R is a sticky receptive tip, and S encloses the ovules. The correct identification of P, Q, R, S is:

- (A) P = stigma, Q = style, R = anther, S = ovary
- (B) P = anther, Q = filament, R = ovary, S = stigma
- (C) P = ovary, Q = filament, R = stigma, S = anther
- (D) P = anther, Q = filament, R = stigma, S = ovary

13. A flower produces abundant pollen but is found to have no stigma, style or ovary of its own, while a second flower on the same plant has a pistil but no stamens. Together these two flowers are best described as:

- (A) two bisexual flowers, since between them they have all parts
- (B) two spore-bearing flowers, since neither makes seeds alone
- (C) unisexual flowers — a male and a female — borne on the same plant
- (D) one fertile and one sterile bisexual flower

14. Tracing the female parts of a flower from the receptive surface down to where seeds will eventually form, the correct order is:

- (A) stigma, then ovary, then style (B) stigma, then style, then ovary
- (C) anther, then filament, then ovary (D) ovary, then style, then stigma

15. Before a bisexual flower opens, a researcher carefully removes its entire ovary while leaving everything else intact, then lets it be pollinated normally. Which later event becomes impossible, and why?

- (A) seed formation, because the ovules that turn into seeds were housed inside the removed ovary
- (B) pollen release, because pollen is manufactured in the ovary
- (C) the stigma receiving pollen, because the stigma sits on the ovary
- (D) the petals wilting, because petals grow out of the ovary

16. A flower has bright petals, healthy stamens, and a pistil whose ovary contains no ovules at all. Even after flawless pollination and a pollen tube reaching the ovary, this flower cannot:

- (A) release pollen, because missing ovules block the anthers
- (B) form seeds, because there are no ovules to develop into seeds after fertilisation
- (C) have its stigma receive pollen, because no ovules means no stigma
- (D) be visited by insects, because they detect the empty ovary

17. A student insists 'the filament is the part on which pollen lands during pollination.' The precise correction is:

- (A) pollen lands on the anther, since that is where it is also stored
- (B) the statement is correct, because the filament is sticky and traps pollen
- (C) pollen lands on the stigma; the filament is only the stalk that holds the anther up
- (D) pollen lands inside the ovary, not on the filament

18. In a maize plant the pollen-shedding tassel sits at the very top while the grain-bearing cob forms lower down on a separate structure. A student concludes maize has a single giant bisexual flower. The correct interpretation is that maize has:

- (A) separate male and female (unisexual) flowers on the same plant
- (B) one bisexual flower spread out over the whole plant
- (C) no flowers at all, reproducing only by spores
- (D) flowers that set grain without any pollen

19. A botanist wishes to collect pure pollen for an experiment and is told to dust a particular flower part. She should target the anther rather than the stigma because:

- (A) the stigma makes the pollen and the anther stores ovules
- (B) both make pollen but the stigma's is too sticky to gather
- (C) neither makes pollen; it must be scraped from the petals
- (D) the anther is where pollen is actually produced, while the stigma only receives it

20. A flower has stamens and a pistil in the same bloom, but its anthers ripen and shed all their pollen days before its own stigma becomes receptive. The most reasonable consequence for this single flower is that:

- (A) its own pollen cannot land on its own ready stigma, favouring pollen from another flower instead
- (B) it cannot be pollinated at all, since its pollen is gone
- (C) it must reproduce by spores because the timing is mismatched
- (D) it switches to vegetative propagation to set seed

21. A bisexual flower is described as 'guaranteed to self-pollinate because it has both anthers and a stigma together.' The most accurate refinement of this statement is:

- (A) the statement is fully correct; both parts always force self-pollination
- (B) the statement is wrong because a single flower can never self-pollinate
- (C) the statement is wrong because bisexual flowers reproduce only by spores
- (D) having both parts makes self-pollination possible, but it is not guaranteed, since timing or structure can still favour cross-pollination

22. Pollen from a mustard flower is carried to the stigma of a different mustard plant, and weeks later seeds form. The pollen-transfer step alone is correctly named:

- (A) cross-pollination
- (B) fertilisation
- (C) self-pollination
- (D) fragmentation

23. A grower wants a tomato flower fertilised only by its own pollen. She bags the unopened flower so no insect or breeze can reach it, and it still sets fruit. This proves the flower underwent:

- (A) self-pollination, using its own pollen on its own stigma
- (B) cross-pollination, since the bag let a neighbour's pollen slip in

- (C) fertilisation without any pollination at all
- (D) spore formation triggered by the darkness inside the bag

24. A young bisexual flower has its anthers snipped off before it opens, is left uncovered near other plants of the same kind, and later sets seed. The pollination responsible must have been:

- (A) self-pollination, using the flower's remaining pollen
- (B) no pollination, since seeds can form without any pollen
- (C) cross-pollination, since the flower's own pollen source was removed
- (D) spore formation by the anther-less flower

25. Two flowers are compared. Flower X is large, scented, brightly coloured and makes sticky pollen; Flower Y is small, drab, scentless and sheds huge amounts of light, dry pollen. Their most likely pollinating agents are:

- (A) Flower X by insects and Flower Y by wind
- (B) Flower X by wind and Flower Y by insects
- (C) both by water currents
- (D) both by insects, the difference being only in size

26. A wind-pollinated grass produces vastly more pollen than an insect-pollinated apple flower of similar size. The deepest reason for this surplus is that:

- (A) wind pollen is heavier, so more grains are needed to make it fly
- (B) insects eat most of the grass pollen before it can land
- (C) the grass deliberately wastes pollen to attract pollinators
- (D) wind scatters pollen at random, so a huge surplus is needed for any grain to chance upon a stigma

27. A flower offers a deep pool of nectar, a sweet scent and broad landing-pad petals. A submerged aquatic flower nearby releases pollen straight into the moving water. The pollinating agents these two sets of features point to are, respectively:

- (A) wind for the nectar-rich flower and insects for the submerged one
- (B) water for both flowers
- (C) insects for both flowers
- (D) insects for the nectar-rich flower and water for the submerged one

28. A poster reads: 'When a bee carries pollen between two flowers, that is the moment the seeds are made.' The precise correction is:

- (A) the poster is right; pollen transfer and seed-making are the same event
- (B) the bee actually fuses the gametes itself while feeding
- (C) the seeds are already finished and the bee merely moves them
- (D) the bee only performs pollination; the seeds form later, after fertilisation inside the ovule

29. Grass flowers are small, dull-green and scentless while apple flowers are showy and fragrant. The simplest explanation for the grass flowers' plain appearance is that they:

- (A) are wind-pollinated, and wind is not drawn by colour or scent
- (B) are pollinated by insects that prefer dull, scentless blooms
- (C) reproduce by spores and so need no pollination
- (D) are pollinated by water seeping over the soil

30. A flower opens, sheds pollen onto its own stigma before any insect arrives, and reliably sets seed even in a season when pollinators are scarce. The chief advantage of this behaviour is that the plant:

- (A) guarantees the widest possible variation among its offspring
- (B) combines the best traits of two different parents
- (C) can reproduce dependably even when no pollinating agent or partner is available
- (D) avoids ever needing to form a zygote

31. Which statement correctly separates self-pollination from cross-pollination?

- (A) self-pollination always needs an insect, while cross-pollination needs none
- (B) self-pollination keeps pollen within the same flower or plant, while cross-pollination carries it to a different plant of the same kind
- (C) self-pollination produces spores while cross-pollination produces seeds
- (D) cross-pollination keeps pollen on the same flower while self-pollination sends it away

32. After pollen settles on a stigma, the male gamete must reach the egg deep inside an ovule. The route it takes is:

- (A) down a pollen tube that grows through the style into the ovary and to the ovule
- (B) along the outside of the filament into the ovary
- (C) blown by the wind directly through the petals to the ovule
- (D) swallowed by the stigma and pushed out through a root

33. Two events are described: (i) pollen lands on a stigma; (ii) the male gamete fuses with the egg in an ovule. The correct relationship is:

- (A) (i) and (ii) are two names for the same event
- (B) (i) is pollination and must occur before (ii), which is fertilisation
- (C) (ii) happens first and triggers (i)
- (D) (i) is fertilisation and (ii) is pollination

34. A flower is visited by bees and its stigma is heavily dusted with pollen, yet weeks later no fruit forms at all. The single most reasonable explanation is that:

- (A) pollination clearly failed, since pollen never reached the stigma
- (B) fertilisation did not occur — the male gamete never fused with the egg in the ovule
- (C) the ovary turned straight into seeds without forming a fruit
- (D) seed dispersal happened before the fruit could develop

35. Immediately after the male gamete fuses with the egg, the very first structure formed is the:

- (A) seed coat
- (B) fruit
- (C) pollen grain
- (D) zygote

36. After fertilisation in a flowering plant, the fates of three structures are traced. The correct matching of ovule, ovary and zygote is:

- (A) ovule becomes the fruit, ovary becomes the seed, zygote becomes the embryo
- (B) ovule becomes the seed, ovary becomes the embryo, zygote becomes the fruit
- (C) ovule becomes the embryo, ovary becomes the fruit, zygote becomes the seed
- (D) ovule becomes the seed, ovary becomes the fruit, zygote becomes the embryo

37. Soon after a pea flower is fertilised, the petals and stamens shrivel and drop, while the base of the flower begins to swell and lengthen into a pod. The swelling, pod-forming part is the:

- (A) anther, which is enlarging into the seeds
- (B) stigma, which is swelling into the fruit
- (C) ovary, which develops into the fruit (the pod)
- (D) filament, which becomes the pod wall

38. A botanist slices open one fruit and counts eight seeds inside. The most direct conclusion about the flower it came from is that its ovary contained:

- (A) eight anthers, one for each seed
- (B) at least eight ovules, each of which became a seed after fertilisation
- (C) a single ovule that later split into eight seeds
- (D) eight stigmas, each of which grew into a seed

39. Arrange these stages of a flowering plant's life in the correct order: (i) the zygote forms; (ii) pollen reaches the stigma; (iii) the seed grows into a new plant; (iv) the ovary ripens into a fruit holding seeds.

- (A) (i), then (ii), then (iii), then (iv)
- (B) (ii), then (i), then (iv), then (iii)
- (C) (ii), then (iv), then (i), then (iii)
- (D) (iv), then (ii), then (i), then (iii)

40. A flower is cross-pollinated, fertilisation occurs, and a fruit forms. Following the fate of one particular ovule, its correct end state is that it becomes a:

- (A) fruit that encloses all the other ovules
- (B) fresh stigma for the plant's next flower
- (C) seed, inside which the zygote it held has grown into an embryo
- (D) pollen grain for the next round of pollination

41. A student claims that as long as a flower is pollinated, a fruit is certain to form. A counter-case that disproves this is a flower that:

- (A) received pollen but in which fertilisation failed, so no fruit developed
- (B) received pollen and promptly formed a fruit, as expected
- (C) was never visited by any insect or breeze
- (D) produced spores instead of pollen from the start

42. A dandelion fruit carries a tuft of fine hairs like a parachute, while a coconut has a thick, air-filled spongy husk. These two structures are adaptations for dispersal by, respectively:

- (A) water for the dandelion and wind for the coconut
- (B) animals for both
- (C) wind for the dandelion and water for the coconut
- (D) explosion for both

43. Walking through tall grass, a hiker finds many small fruits with stiff hooks clinging to her socks; in a nearby orchard, birds swallow sweet berries and later void the hard seeds far away. These two cases both illustrate dispersal by:

- (A) wind in both cases, the hooks and berries merely catching breezes
- (B) animals — by clinging to fur or clothing in one case and by being eaten and passed in the other
- (C) water in both cases, since animals carry the fruits to streams
- (D) explosion in both cases, triggered by the hooks and the eating

44. A ripe pod dries in the sun and suddenly snaps open, flinging its seeds a short distance around the parent. The seeds bear no hooks, hairs or buoyant husk. The dispersal agent here is:

- (A) wind, which must catch the seeds as they leave the pod
- (B) animals, which carry the closed pod away first
- (C) the bursting of the dried fruit itself (explosion), needing no outside carrier
- (D) water, which washes the seeds from the split pod

45. Why are juicy, brightly coloured, sweet fruits regarded as an adaptation for animal dispersal rather than for wind or water?

- (A) the bright colour makes the fruit light enough to be blown by wind
- (B) animals are drawn to eat the fruit and then drop or pass the tough seeds far from the parent
- (C) the juice helps the whole fruit float away on rivers
- (D) the sweetness makes the fruit burst and fling its seeds

46. A plant always drops its seeds straight down at its own base, with no structure aiding travel. Over the long term, the most likely disadvantage of this is that:

- (A) the seedlings cannot photosynthesise when close to the parent
- (B) the seeds turn back into spores if they land near the parent
- (C) the seedlings get far too much space and starve
- (D) the crowded seedlings compete with the parent and one another for light, water, minerals and space

47. Beyond easing overcrowding near the parent, what further benefit does seed dispersal give a plant species?

- (A) it makes each seed heavier so it stays in one place
- (B) it speeds up pollination of the parent flower
- (C) it lets the species reach and colonise new areas where conditions may be favourable
- (D) it removes the need for fertilisation in the next generation

48. A drumstick seed has papery wings, a coconut a spongy buoyant husk, and a Xanthium fruit tiny hooks. A student says all three are adapted to the same dispersal agent. The correct response is:

- (A) they are adapted to different agents — wind, water and animals respectively
- (B) they are all adapted to wind, the husk and hooks acting as sails
- (C) they are all adapted to water, since each can float
- (D) they are all adapted to explosion when ripe

49. Which seed feature would be LEAST helpful for dispersal by wind?

- (A) a heavy seed with a thick, stony coat
- (B) a tuft of fine hairs forming a parachute
- (C) thin, flat papery wings
- (D) a very small, light body

50. A cultivated banana variety bears no usable seeds, yet new banana plants keep appearing in the field. The best explanation is that the banana is propagated:

- (A) by spores released from the ripe fruit
- (B) by tiny wind-blown seeds too small to notice
- (C) vegetatively, from its underground stem, so no seeds are needed
- (D) by cross-pollination that hides the seeds inside the pulp

51. Which single observation would most convincingly show that a new plant arose by sexual reproduction rather than by vegetative propagation?

- (A) it grew from a buried piece of underground stem
- (B) it sprouted from a bud on a leaf margin
- (C) it rooted from a stem cutting placed in water
- (D) it grew from a seed that had formed inside a fruit after a flower was fertilised

52. A spore and a seed are often confused. The most fundamental difference between them is that:

- (A) a spore forms after fertilisation while a seed needs no gametes
- (B) a seed forms after fertilisation (fusion of gametes), whereas a spore is an asexual unit formed without any gamete fusion
- (C) both form only after pollination by insects
- (D) a seed is always lighter and more wind-blown than a spore

53. A student classifies four cases of new-plant formation: Bryophyllum leaf-margin buds, ginger from its underground stem, yeast budding off new cells, and a mango grown from its seed. The one that does NOT belong with the rest is:

- (A) the mango, because it forms by sexual reproduction while the other three are asexual
- (B) the yeast bud, because budding is the only sexual process listed
- (C) the ginger stem, because underground parts cannot reproduce
- (D) the Bryophyllum buds, because leaves never make new plants

54. Hydra, a small water animal, grows a bulge that enlarges and detaches as a new individual. The plant-kingdom organism whose reproduction works by the same basic process is:

- (A) Spirogyra, which breaks into fragments
- (B) a fern, which scatters spores
- (C) yeast, which buds off new cells
- (D) a potato, which sprouts from buds in its eyes

55. Why must the step an insect performs when carrying pollen between two flowers be called pollination and not fertilisation?

- (A) the insect fuses the gametes itself while sipping nectar
- (B) the insect carries finished seeds, not pollen, between flowers
- (C) fertilisation is simply another word for any insect visit
- (D) the insect only moves pollen to the stigma; the actual fusion of gametes happens later inside the ovule

56. A farmer wants every new plant of a prized seedless variety to be genetically identical to that one parent, and quickly. The method that meets both aims is:

- (A) collecting and sowing the parent's seeds
- (B) planting stem cuttings taken from that parent
- (C) letting the wind cross-pollinate it with neighbours
- (D) waiting for the variety to form spores

57. A student writes: 'All reproduction in plants must involve a flower.' The single clearest counter-example showing this is false is:

- (A) a mango tree, which flowers before it fruits
- (B) a fern, which makes new ferns from spores and never produces a flower
- (C) an insect-pollinated rose, whose bright flower attracts bees
- (D) a papaya, which bears separate male and female flowers

58. Two beds of one crop are planted: Bed A from seeds and Bed B from cuttings of a single parent. After several seasons a grower notes that Bed A shows a few plants with a new, useful trait the parent never had, while Bed B never does. The reason is that:

- (A) seeds arise from fertilisation, which reshuffles traits and can yield variation, while cuttings only clone the parent
- (B) cuttings produce spores that erase the parent's traits each generation
- (C) seeds skip fertilisation and so invent traits from nothing
- (D) cuttings need two parents while seeds need only one

59. A gardener bends a low, still-attached rose branch to the ground, buries part of it in soil until roots form, then cuts the rooted portion free as a new plant. This technique is best described as:

- (A) fragmentation, because the branch was eventually cut
- (B) cross-pollination between the branch and the soil
- (C) spore formation triggered by burying the stem
- (D) layering, a form of vegetative propagation

60. A teacher lists how four plants are usually raised: a sugarcane plant from a stem piece, a sweet-potato plant from a root, a Bryophyllum plant from a leaf, and a marigold from its seed. A student is asked which plant part used here is NOT a route of vegetative propagation. The correct answer is:

- (A) the sugarcane's stem, because stems cannot grow new plants
- (B) the sweet-potato's root, because roots never sprout shoots
- (C) the Bryophyllum's leaf, because leaves cannot bear buds
- (D) the marigold's seed, because a seed forms after fertilisation and is not a vegetative part

Solutions — Science (Class 7), Chapter 12:

Reproduction in Plants — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	D	21	D	31	B	41	A	51	D
2	B	12	D	22	A	32	A	42	C	52	B
3	B	13	C	23	A	33	B	43	B	53	A
4	C	14	B	24	C	34	B	44	C	54	C
5	A	15	A	25	A	35	D	45	B	55	D
6	C	16	B	26	D	36	D	46	D	56	B
7	B	17	C	27	D	37	C	47	C	57	B
8	C	18	A	28	D	38	B	48	A	58	A
9	C	19	D	29	A	39	B	49	A	59	D
10	A	20	A	30	C	40	C	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (D)** Vegetative propagation copies one parent's material without any fusion of gametes, so every offspring is a clone; without the reshuffling that comes from combining two parents' contributions, the cutting route cannot itself introduce new traits, regardless of how seedless the fruit is.
- (B)** Seeds form from two parents' gametes, so their offspring vary; that variation means some individuals chance to resist a new threat. The cuttings are identical copies, so a weakness in the parent is shared by every plant, allowing uniform devastation.
- (B)** A potato's eyes are buds, and shoots grow from those buds; gouging them out leaves the stored food intact but removes the growth points, so that piece cannot send up new shoots even though it has plenty of food.
- (C)** Bryophyllum's leaf margins carry buds that each grow into a new plant — a vegetative, asexual route. The fern's brown patches are spore cases; its young plants come

from spores. So one is bud-based vegetative propagation and the other is spore-based, which are genuinely different mechanisms.

5. (A) Budding is the pinching-off of a single outgrowth from a parent, which is exactly what the yeast shows. Fragmentation is the breaking of the body into pieces that each regrow, which is what Spirogyra shows. So only the alga's case is fragmentation.

6. (C) The mushroom, moss and fern are united by reproducing through spores. The potato instead grows new plants from the buds (eyes) of its underground stem, a vegetative route, so it is the one that does not share the grouping feature.

7. (B) A spore is shielded in a tough case and stays dormant through unfavourable conditions, then germinates when it turns moist. It needs no partner and does not rush to germinate; that ability to wait out a drought is its key advantage over constant division.

8. (C) Money plants root readily from stem cuttings, giving a quick, reliable clonal supply. Breeding trials need variation, which comes from seeds produced by fertilisation. Matching each goal to the right route means cuttings for the money plant and seeds for the tomatoes.

9. (C) Grafting is a vegetative method: the joined shoot grows on as a copy of the chosen variety, so its fruit shows that variety's traits. There is no fusion of gametes, so the fruit is not a new blend; the rootstock mainly supplies roots.

10. (A) Asexual reproduction is defined by the absence of gamete fusion, not by the presence of buds. Bulbs and underground stems are vegetative (asexual) routes, and spores are asexual units too, so all three examples are asexual.

11. (D) Both cuttings come from one parent with no gamete fusion, so both are genetically identical clones. The rooting medium can affect how well each grows but does not change the inherited material, so the two roses are not genetically different.

12. (D) Pollen is produced in the anther, held up by the filament; the sticky receptive tip is the stigma, and the ovules sit inside the ovary. Matching each clue to those parts gives anther, filament, stigma, ovary in that order.

13. (C) Each flower carries only one set of reproductive parts — one only stamens, the other only a pistil — so each is unisexual (a male and a female flower). They are not bisexual, because a bisexual flower must carry both sets within the same bloom.

14. (B) The receptive surface is the stigma; below it lies the style, and at the base sits the ovary that holds the ovules destined to become seeds. So the descent runs stigma to style to ovary.

- 15. (A)** Seeds develop from ovules, and the ovules are inside the ovary; removing the ovary removes the ovules. Pollen is still made in the anthers and the stigma can still receive pollen, but with no ovules there can be no seeds.
- 16. (B)** Seeds come from ovules; with no ovules present, fertilisation cannot lead to any seeds. Pollen release, stigma reception and insect visits all depend on other parts that are intact, so only seed formation is blocked.
- 17. (C)** During pollination pollen is deposited on the stigma, the receptive tip of the pistil. The filament merely supports the anther; it is not the landing surface, and the anther makes pollen rather than receiving it.
- 18. (A)** The tassel carries male flowers that shed pollen and the cob carries female flowers that become grain. Since the male and female parts are in separate flowers, maize bears unisexual flowers on one plant, not a single bisexual flower.
- 19. (D)** Pollen is manufactured in the anther, so that is the source to dust for a pure sample. The stigma is merely the receiving surface; it does not produce pollen, so collecting from it would not yield the grains she needs.
- 20. (A)** If the anthers shed pollen before the stigma is ready, the flower's own pollen has dispersed by the time its stigma can receive any. So self-pollination is hindered and pollen arriving from another flower of the same kind is what can reach the receptive stigma.
- 21. (D)** Carrying both anthers and a stigma in one bloom only makes self-pollination possible. Whether it actually happens depends on factors like the timing of pollen release versus stigma readiness, so self-pollination is enabled but not guaranteed.
- 22. (A)** Moving pollen to the stigma of a different plant of the same kind is cross-pollination. The later seed formation depends on fertilisation, a separate fusion step, so the transfer itself is just cross-pollination.
- 23. (A)** With the bag sealing out all outside pollen, the only source left is the flower's own anthers. Fruit set therefore means its own pollen reached its own stigma — self-pollination — followed by fertilisation.
- 24. (C)** Removing the anthers strips the flower of its own pollen, so any pollen reaching its stigma must come from another plant; that is cross-pollination. Seeds cannot form without pollination, so the seed set confirms pollen arrived from elsewhere.
- 25. (A)** Colour, scent and sticky pollen attract and cling to insects, so X is insect-pollinated. Light, dry, abundant pollen and the lack of showy features suit wind carriage, so Y is wind-pollinated.

26. (D) Wind delivers pollen with no aim, so most grains miss any stigma; producing an enormous surplus raises the odds that a few will land where they should. Insects, by contrast, ferry pollen more directly, so far less is required.

27. (D) Nectar, scent and landing-pad petals are adaptations to attract insects. Releasing pollen into flowing water so it is carried to other flowers is an adaptation for water pollination. So the two flowers are pollinated by insects and water respectively.

28. (D) An insect's job is pollination — moving pollen to a stigma. Seeds form only after fertilisation, when the male gamete fuses with the egg in the ovule, an event that happens after the pollen lands, not during the bee's visit.

29. (A) Showy colour and scent are investments that pay off only when pollinators must be attracted. Wind responds to neither, so wind-pollinated grasses gain nothing from bright petals or fragrance and remain plain.

30. (C) Self-pollination lets the plant set seed without relying on an outside agent or another plant, so reproduction is secure even when pollinators are absent. It does not maximise variation — that is the strength of cross-pollination — and it still requires a zygote.

31. (B) The distinction is about destination: self-pollination delivers pollen within the same flower or plant, whereas cross-pollination delivers it to a different plant of the same kind. Both can yield seeds, and neither necessarily requires an insect.

32. (A) The pollen grain grows a pollen tube that pushes down through the style and into the ovary, carrying the male gamete to the ovule where fusion occurs. There is no path along the filament, through petals, or via the roots.

33. (B) Pollen landing on the stigma is pollination; the fusion of the male gamete with the egg is fertilisation. Fertilisation cannot occur until pollen has arrived and a tube has grown, so pollination must come first.

34. (B) The dusted stigma shows pollination succeeded, so the breakdown must lie later. With no fruit, fertilisation did not take place: the male gamete never fused with the egg, so no fruit could develop from the ovary.

35. (D) Fusion of the male gamete and the egg produces a single cell, the zygote, which later develops into the embryo. The seed coat and fruit form afterward from other parts, and the pollen grain existed before fusion.

36. (D) The ovule matures into the seed, the surrounding ovary swells into the fruit, and the zygote within grows into the embryo. Confusing any pair — for example swapping seed and fruit — is the classic error this checks.

37. (C) The pod is the fruit of the pea, and fruits develop from the ovary. As the ovary ripens into the pod, the now-redundant petals, stamens and stigma wither and fall, leaving the swelling ovary behind.

38. (B) Each seed develops from one fertilised ovule, so eight seeds imply the ovary held at least eight ovules. Anthers and stigmas do not become seeds, and a single ovule yields a single seed rather than splitting into many.

39. (B) Pollen first reaches the stigma, then fertilisation forms the zygote, then the ovary ripens into a seed-bearing fruit, and finally a seed germinates into a new plant. So the order is pollination, zygote, fruit, new plant.

40. (C) An ovule matures into a seed, and the zygote it contained develops inside it into the embryo. The fruit forms from the whole ovary, not from a single ovule, and ovules never turn into stigmas or pollen.

41. (A) Pollination only delivers pollen; a fruit forms only after fertilisation occurs. A flower that is pollinated but in which the gametes never fuse sets no fruit, directly disproving the claim that pollination alone guarantees fruit.

42. (C) Fine hairs catch the wind and keep a light fruit aloft, so the dandelion is wind-dispersed. A buoyant, air-filled husk lets the coconut float and ride currents, so it is water-dispersed. Each structure is matched to a different agent.

43. (B) Hooks that catch on fur or clothing and sweet fruits that are eaten and the seeds later dropped are both ways animals carry seeds away. Both scenarios are therefore animal dispersal, by two different mechanisms.

44. (C) The sudden splitting of the dried pod throws the seeds out by itself; the lack of hooks, hairs or a buoyant husk rules out animals, wind and water as the means. This self-powered scattering is dispersal by explosion.

45. (B) Colour and sweetness lure animals to eat the fruit; the hard seeds survive and are deposited elsewhere, carrying the plant to new ground. Juiciness adds weight rather than aiding flight, and does not make fruits float or burst.

46. (D) Seeds landing in a dense cluster at the parent's base must share the same light, water, minerals and space, so they compete heavily. Dispersal would relieve this crowding; the other options invent effects that do not occur.

47. (C) Carrying seeds away spreads the species to fresh ground that may suit it, increasing its chances of establishing new populations. Dispersal does not add weight, hasten pollination, or replace the need for fertilisation.

48. (A) Papery wings catch the wind, a buoyant husk floats on water, and hooks cling to animal fur — three different structures for three different agents. So the three seeds are not adapted to one common agent.

49. (A) Wind dispersal favours lightness and surfaces that catch air, such as hairs, wings or a tiny body. A heavy seed with a thick stony coat resists being lifted or carried, so that feature is the least useful for wind.

50. (C) Lacking usable seeds, this banana cannot rely on seed reproduction; new plants arise vegetatively from its underground stem. That clonal route needs no seeds, explaining how the variety keeps spreading despite being seedless.

51. (D) A seed formed inside a fruit after fertilisation is the product of fusing gametes — the hallmark of sexual reproduction. Underground stems, leaf-margin buds and cuttings are all vegetative routes that involve no gamete fusion.

52. (B) A seed is the outcome of fertilisation, the fusion of male and female gametes. A spore is produced asexually with no such fusion. That presence or absence of gamete fusion is the deepest distinction between the two.

53. (A) Leaf-margin buds, underground-stem growth and yeast budding are all asexual, with no gamete fusion. The mango seed forms after fertilisation, so it alone is sexual reproduction and is the odd one out.

54. (C) Hydra reproduces by budding — an outgrowth swells and separates. Among the options, yeast does the same, budding off new cells. Spirogyra fragments, the fern uses spores, and the potato's eyes are buds on a stem, not a budding-off of the whole organism.

55. (D) An insect's role is limited to transferring pollen to a stigma, which is pollination. Fertilisation — the fusion of the male gamete with the egg — occurs afterward inside the ovule and is not something the insect performs.

56. (B) Stem cuttings are a vegetative, clonal route, so every new plant copies the single parent exactly and grows fast. Seeds and cross-pollination introduce variation, and the variety does not reproduce by spores, so those routes fail the identical-copy goal.

57. (B) A fern reproduces by spores and never bears a flower at all, directly disproving the claim. The mango, rose and papaya all do flower, so they cannot serve as counter-examples to a statement about flowers being essential.

58. (A) Seed plants come from the fusion of two parents' gametes, which mixes traits and can throw up new combinations. Cuttings are clones with no such mixing, so Bed B stays identical to the parent while Bed A can vary.

59. (D) Encouraging roots on a branch while it is still joined to the parent, then severing it, is layering — a vegetative method producing a clone. It is not fragmentation, which is a body breaking into self-sustaining pieces, nor anything to do with pollen or spores.

60. (D) Vegetative propagation grows new plants from a stem, root or leaf of one parent without gamete fusion, so the sugarcane stem, sweet-potato root and Bryophyllum leaf all qualify. A seed, however, is the product of fertilisation, making it the non-vegetative route in the list.